

Industrial Machine Maintenance and Spare-Parts Analysis Summary

1. High-Level Maintenance Metrics Overview

This analysis evaluates three years of daily sensor data and spare-parts transactions from 3 January 2022 to 1 January 2025. The scope encompasses 10 assets across three plants (PUN-01, CHN-02, and DHR-03). The core objective is to identify systemic failure patterns, quantify maintenance expenditure, and isolate high-fidelity early-warning signals to transition from a reactive to a predictive reliability model.

Metric	Value	Total
Spare-Parts Spend	₹1,33,40,945 (₹1.33 Crore)	
Total Work Orders Raised	42,793	
Breakdown (Unplanned) Work Orders	21,636 (50.6%)	
Planned Maintenance Work Orders	21,157 (49.4%)	

Top Takeaways

- Unstable Maintenance State:** With a breakdown rate of 50.6%, the fleet is in a highly reactive state. A nearly 1:1 ratio of unplanned to planned work indicates that current preventive schedules are failing to intercept faults before they reach functional failure.
- Pareto Drivers:** Bearings and Seals & Gaskets represent nearly 50% of total spend and are the leading contributors to breakdown counts. These two categories are the primary levers for both cost reduction and uptime improvement.
- Predictive Signal Dominance:** Power consumption and load percentages show clear, statistically significant deviations prior to failure, whereas traditional temperature and vibration metrics remain relatively flat until the point of breakdown.

2. Spare-Parts Spend Analysis

2.1. Expenditure by Part Family and Individual Parts

- Highest-Spend Part Family: Bearings** (₹32.60 Lakh, representing 24.4% of total spend).
- Costliest Single Part: Hydraulic Seal Kit 50mm Bore** (₹28.48 Lakh). This is a critical outlier; the spend on this single SKU is more than double the next most expensive item in the inventory, suggesting a significant opportunity for procurement negotiation or engineering substitution.
- Primary Drivers:** Bearings and Seals & Gaskets together account for approximately 47% of the total ₹1.33 Crore expenditure.

2.2. Spend by Machine Type and Asset Criticality

Budget allocation is distributed evenly across machine types, with no single machine class disproportionately consuming the maintenance budget.

Machine Type	Spend (₹)	% of Total
Belt Conveyor	27,45,522	20.6%
Screw Compressor	27,14,335	20.3%
EOT Crane	26,42,723	19.8%
Hydraulic Press	26,32,803	19.7%
CNC Lathe	26,05,562	19.5%

Asset Criticality Analysis The data confirms a healthy **spend-to-criticality alignment**. Criticality-A assets account for ₹68.87 Lakh (51.6% of total spend), ensuring the most vital equipment receiving the highest resource allocation. Criticality-B and C assets account for 37.0% and 11.4% of the spend, respectively.

3. Breakdown and Reliability Statistics

3.1. Machine Type and Plant Performance

- **High-Risk Machine Types:** The Hydraulic Press and Screw Compressor exhibit the highest breakdown work-order counts and should be prioritized for enhanced condition monitoring.
- **Plant Performance Outlier:** Plant **DHR-03** reports a concerning breakdown rate of 69.0%. This is highly inefficient compared to PUN-01; DHR-03 generates significantly higher breakdown density despite having 25% fewer assets (3 machines vs. 4 in PUN-01).

3.2. Criticality and Part Family Breakdown Counts

A technical red flag exists in the criticality distribution: Criticality-B assets show a higher percentage of breakdown events (43.3%) than Criticality-A assets (38.8%). This suggests either a misclassification of asset criticality or a neglect of "B" class assets that are suffering from under-maintenance.

Part Family	Breakdown Work Orders	Share (%)
Bearing	4,221	19.5%
Seal & Gasket	3,118	14.4%
Sensor	2,454	11.3%
Electrical	2,417	11.2%
Filter	2,136	9.9%
Drive Belt	2,134	9.9%
Coupling	1,987	9.2%
Fastener	1,626	7.5%
Lubrication	1,543	7.1%

3.3. Asset-Specific Reliability

The following five assets are chronically unreliable, accounting for over 50% of all breakdown events across the 10-machine fleet:

- **AST-2017** (2,647 breakdowns)
- **AST-5003** (2,565 breakdowns)
- **AST-4055** (2,558 breakdowns)
- **AST-2031** (2,311 breakdowns)
- **AST-4061** (2,102 breakdowns)

4. Sensor-Based Early Warning Signals

The delta between normal operations and breakdown events provides the foundation for an automated alert logic.

Sensor	Normal Days (avg)	Breakdown Days (avg)	% Change
Power Consumption (kW)	26.13	29.40	+12.5%
Load (%)	67.92	72.18	+6.3%
Vertical Vibration (mm/s)	2.89	3.02	+4.5%
Horizontal Vibration (mm/s)	3.85	4.02	+4.4%

Engineering Synthesis

- **Leading Indicators:** Power consumption (+12.5%) is the most sensitive leading indicator of impending failure. When paired with load percentage (+6.3%), it forms a robust two-variable alert trigger for preemptive inspection.
- **Weaker Predictors:** Within this specific fleet, temperature and vibration are poor standalone predictors. Temperature shifts are marginal (<1%), and vibration increases remain under 5% prior to failure, making them late-stage indicators rather than early-warning signals.

5. Strategic Recommendations

5.1. Immediate Actions

1. **Reliability Review of Top 5 Assets:** Initiate immediate root-cause analysis (RCA) or replacement feasibility studies for AST-2017, AST-5003, AST-4055, AST-2031, and AST-4061.
2. **Power/Load Monitoring Alerts:** Deploy automated SCADA/IoT alerts for sustained increases in power draw and load percentage.
3. **Bearing Replacement Standardization:** As Bearings are the #1 cause of breakdowns (19.5%) and the top spend category, moving from "run-to-fail" to a fixed-interval replacement or condition-based strategy will provide the highest immediate ROI.

5.2. Medium-Term and Process Improvements

- **Maintenance Mix Shift:** Implement a target to move 10–15% of current breakdown work orders into the planned maintenance category to stabilize operations and reduce emergency procurement premiums.
- **DHR-03 Audit:** Conduct a site audit at DHR-03 to identify why its 3-machine fleet is significantly underperforming compared to the larger fleet at PUN-01.
- **Criticality-B Investigation:** Review the maintenance strategy for Criticality-B assets to address the 43.3% breakdown share.
- **Enhanced Data Tracking:**
 - Integrate a 'load-power ratio' metric to increase the sensitivity of early-warning alerts.
 - Institutionalize **Mean-Time-Between-Failures (MTBF)** tracking to measure the effectiveness of these reliability interventions over time.